VM Setup Commands Guide: vProfile Stack (5 VMs)

Manual setup commands for each VM, organized so you can follow them one VM at a time.

1. Overview

VM	Role	Main software	Port(s)	Package manager
db	Database	MariaDB	3306	yum
memcache	Cache	Memcached	11211 (TCP), 11111 (UDP)	yum
rabbitmq	Message broker	RabbitMQ 3.8	5672	dnf
tomcat	Application server	Java 11, Tomcat 9.0.75, Maven 3.9.9	8080	dnf
nginx	Web server / reverse proxy	nginx	80, 443	dnf

2. Before you start

- Run each section's commands **on that VM only**, not on the others.
- The config files use the names `db`, `memcache`, `rabbitmq` and `tomcat`, so every VM must be able to resolve these names (DNS or `/etc/hosts`).
- Recommended order: db, memcache, rabbitmq, tomcat, nginx. The backend services should be running before you start the app.
- Lines starting with `#` are comments, not commands.
- All passwords in this guide are lab values only.

3. DB VM (MariaDB)

Purpose: stores the application data in the `accounts` database.

3.1 Install packages

```
sudo yum update -y
sudo yum install epel-release -y
sudo yum install -y mariadb-server
sudo yum install -y git
```

3.2 Start the service

```
sudo systemctl start mariadb
sudo systemctl enable --now mariadb
```

3.3 Secure the server and create the app database and user

Open the MariaDB shell:

```
sudo mysql -u root -proot accounts
```

Then run these SQL statements inside the shell:

```
ALTER USER 'root'@'localhost' IDENTIFIED BY 'root';
DROP USER IF EXISTS ''@'localhost';
DROP USER IF EXISTS ''@'$(hostname)';
DROP USER IF EXISTS 'root'@'%';
DROP DATABASE IF EXISTS test;
DELETE FROM mysql.db WHERE Db='test' OR Db='test\\_%';
CREATE DATABASE IF NOT EXISTS accounts;
CREATE USER IF NOT EXISTS 'admin'@'%' IDENTIFIED BY 'admin123';
GRANT ALL PRIVILEGES ON accounts.* TO 'admin'@'%';
FLUSH PRIVILEGES;
```

3.4 Import the application data

```
sudo mkdir -p /opt/vprofile-project
sudo chmod -R 755 /opt/vprofile-project
cd /opt/vprofile-project
sudo git clone -b main https://github.com/hkhcoder/vprofile-project.git
cd vprofile-project
sudo mysql -u root -proot accounts < src/main/resources/db_backup.sql
```

3.5 Test the connection and tables (optional)

```
mysql -u root -p accounts
```

```
show tables;
exit;
```

3.6 Restart and open the firewall

```
sudo systemctl restart mariadb
sudo systemctl disable firewalld
sudo firewall-cmd --zone=public --add-port=3306/tcp --permanent
sudo firewall-cmd --reload
sudo systemctl restart mariadb
```

4. Memcache VM (Memcached)

Purpose: caches data for the application. Listens on port 11211.

4.1 Install and start

```
sudo yum update -y
sudo yum install epel-release -y
sudo yum install memcached -y
sudo systemctl start memcached
sudo systemctl enable memcached
sudo systemctl status memcached
```

4.2 Allow connections from other VMs

Change the listen address from `127.0.0.1` to `0.0.0.0`:

```
sudo sed -i 's/127.0.0.1/0.0.0.0/g' /etc/sysconfig/memcached
sudo systemctl restart memcached
```

4.3 Open the firewall

```
sudo systemctl start firewalld
sudo systemctl enable firewalld
sudo firewall-cmd --add-port=11211/tcp
sudo firewall-cmd --runtime-to-permanent
sudo firewall-cmd --add-port=11111/udp
sudo firewall-cmd --runtime-to-permanent
```

4.4 Start Memcached with TCP and UDP ports

```
sudo memcached -p 11211 -U 11111 -u memcached -d
```

5. RabbitMQ VM

Purpose: message broker. Listens on port 5672.

5.1 Install packages

```
sudo yum update -y
sudo yum install epel-release -y
sudo yum install wget -y
cd /tmp/
sudo dnf -y install centos-release-rabbitmq-38
sudo dnf --enablerepo=centos-rabbitmq-38 -y install rabbitmq-server
```

5.2 Start the service

```
sudo systemctl enable --now rabbitmq-server
```

5.3 Allow remote logins and create the app user

```
sudo sh -c 'echo "[{rabbit, [{loopback_users, []}]}]." >
/etc/rabbitmq/rabbitmq.config'
sudo rabbitmqctl add_user test test
sudo rabbitmqctl set_user_tags test administrator
sudo rabbitmqctl set_permissions -p "/" test ".*" ".*" ".*"
sudo systemctl restart rabbitmq-server
```

5.4 Open the firewall and make sure the service starts on boot

```
sudo systemctl start firewalld
sudo systemctl enable firewalld
sudo firewall-cmd --add-port=5672/tcp
sudo firewall-cmd --runtime-to-permanent
sudo systemctl start rabbitmq-server
sudo systemctl enable rabbitmq-server
```

6. Tomcat VM (application server)

Purpose: runs the Java application. Listens on port 8080.

6.1 Install Java and tools

```
sudo dnf update -y
sudo dnf install epel-release -y
sudo dnf -y install java-11-openjdk java-11-openjdk-devel
sudo dnf install git wget unzip zip -y
sudo dnf install git maven wget -y
```

6.2 Download and install Tomcat

```
cd /tmp/
sudo wget "https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.75/bin/apache-tomcat-9.0.75.tar.gz"
sudo tar xzvf apache-tomcat-9.0.75.tar.gz
sudo useradd --home-dir /usr/local/tomcat --shell /sbin/nologin tomcat
sudo cp -r /tmp/apache-tomcat-9.0.75/* /usr/local/tomcat/
sudo chown -R tomcat.tomcat /usr/local/tomcat
sudo chmod +x /usr/local/tomcat/bin/*.sh
```

6.3 Create the systemd service

Remove any old service file, then create `/etc/systemd/system/tomcat.service` with this content:

```
sudo rm -rf /etc/systemd/system/tomcat.service
sudo cat /etc/systemd/system/tomcat.service
```

```
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=simple
User=tomcat
Group=tomcat
WorkingDirectory=/usr/local/tomcat
Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk"
Environment="CATALINA_PID=/usr/local/tomcat/temp/tomcat.pid"
Environment="CATALINA_HOME=/usr/local/tomcat"
Environment="CATALINA_BASE=/usr/local/tomcat"
ExecStart=/usr/local/tomcat/bin/catalina.sh run
ExecStop=/usr/local/tomcat/bin/shutdown.sh
RestartSec=10
Restart=always

[Install]
WantedBy=multi-user.target
```

Then load and start it:

```
sudo systemctl daemon-reload
sudo systemctl start tomcat
sudo systemctl enable tomcat
```

6.4 Install Maven 3.9.9

```
cd /tmp/
sudo wget https://archive.apache.org/dist/maven/maven-3/3.9.9/binaries/apache-maven-3.9.9-bin.zip
sudo unzip apache-maven-3.9.9-bin.zip
sudo cp -r apache-maven-3.9.9 /usr/local/maven3.9
sudo chown tomcat.tomcat /usr/local/maven3.9 -R
sudo ln -sf /usr/local/maven3.9/bin/mvn /usr/bin/mvn
```

6.5 Get the application source

```
sudo mkdir -p /opt/vprofile-project
sudo chmod -R 755 /opt/vprofile-project
cd /opt/vprofile-project
sudo git clone -b main https://github.com/hkhcoder/vprofile-project.git
cd vprofile-project
```

6.6 Replace the application configuration

Remove the old file, then create `/opt/vprofile-project/vprofile-project/src/main/resources/application.properties` with this content:

```
sudo rm -rf /opt/vprofile-project/vprofile-project/src/main/resources/application.properties

sudo cat /opt/vprofile-project/vprofile-project/src/main/resources/application.properties
```

```
#JDBC Configutation for Database Connection
jdbc.driverClassName=com.mysql.cj.jdbc.Driver
jdbc.url=jdbc:mysql://db:3306/accounts?useUnicode=true&characterEncoding=UTF-8&zeroDateTimeBehavior=convertToNull
jdbc.username=admin
jdbc.password=admin123

#Memcached Configuration
memcached.active.host=memcache
memcached.active.port=11211

#For StandBy Host
memcached.standBy.host=127.0.0.2
memcached.standBy.port=11211

#RabbitMq Configuration
rabbitmq.address=rabbitmq
rabbitmq.port=5672
rabbitmq.username=test
```

```
rabbitmq.password=test

#Elasticsearch Configuration
elasticsearch.host=localhost
elasticsearch.port=9300
elasticsearch.cluster=vprofile
elasticsearch.node=vprofilenode

spring.servlet.multipart.max-file-size=128KB
spring.servlet.multipart.max-request-size=128KB
logging.level.org.springframework.security=DEBUG

# application.properties
spring.security.user.name=admin_vp
spring.security.user.password=admin_vp
spring.security.user.roles=ADMIN
spring.mvc.view.prefix=/WEB-INF/views/
spring.mvc.view.suffix=.jsp

# Hibernate SQL Queries
spring.jpa.show-sql=false
spring.jpa.properties.hibernate.format_sql=false
logging.level.org.hibernate.SQL=OFF
logging.level.org.hibernate.type=OFF

# Debug Logging for SecurityServiceImpl
logging.level.com.visualpathit.account.service.SecurityServiceImpl=OFF
```

6.7 Build and deploy

```
sudo /usr/local/maven3.9/bin/mvn install
sudo systemctl stop tomcat
sudo rm -rf /usr/local/tomcat/webapps/ROOT
sudo rm -rf /usr/local/tomcat/webapps/ROOT.war
sudo cp target/vprofile-v2.war /usr/local/tomcat/webapps/ROOT.war
sudo chown tomcat.tomcat /usr/local/tomcat/webapps -R
sudo systemctl start tomcat
```

6.8 Firewall and restart

```
sudo systemctl stop firewalld
sudo systemctl disable firewalld
sudo systemctl restart tomcat
```

7. Nginx VM (reverse proxy)

Purpose: receives user traffic on port 80 and forwards it to Tomcat.

7.1 Install and start

```
sudo dnf update -y
sudo dnf upgrade -y
sudo dnf install -y nginx
sudo systemctl start nginx
sudo systemctl enable --now nginx
```

7.2 Configure the proxy

Create `/etc/nginx/conf.d/vproapp.conf` with this content:

```
upstream vproapp {
    server tomcat:8080;
}

server {
    listen 80 default_server;
    server_name _;

    location / {
        proxy_pass http://vproapp;
    }
}
```

7.3 Open the firewall and allow nginx to reach Tomcat

```
sudo systemctl enable --now firewalld
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --permanent --add-service=https
sudo firewall-cmd --reload
sudo setsebool -P httpd_can_network_connect 1
sudo systemctl restart nginx
```

8. Quick checks after setup

VM	Check	Expected result
db	`sudo mysql -u root -proot accounts -e 'show tables;'`	The application tables are listed
memcache	`sudo systemctl status memcached`	`active (running)`
rabbitmq	`sudo rabbitmqctl list_users`	User `test` with the `administrator` tag
tomcat	`sudo systemctl status tomcat` and `curl -I http://localhost:8080`	Service running and an HTTP response
nginx	`curl -I http://localhost`	An HTTP response from the application

Final test: open `http://<nginx-vm-ip>` in a browser.

9. Notes and possible fixes

These points came up while organizing the commands. Check them if something fails.

- **First MariaDB login (3.3):** the command uses `-proot` before the root password has been set. On a fresh install, use `sudo mysql` for the first login, then run the `ALTER USER` statement.
- **`$(hostname)` in SQL (3.3):** the shell does not expand it inside the MariaDB prompt. Replace it with the real host name, or skip that line.

- **Firewall on db and tomcat:** `systemctl disable firewalld` only stops it starting at boot. Rules added with `firewall-cmd` while it is running are lost on reboot unless firewalld is enabled. Choose one approach: keep firewalld on with the needed ports open, or turn it off completely in a lab.
- **Memcached (4.4):** the last command starts a second Memcached process next to the systemd service. If port 11211 is already in use, either stop the service first or edit `/etc/sysconfig/memcached` to include the UDP option instead.
- **Maven on Tomcat (6.1 and 6.4):** Maven is installed twice, once with `dnf` and once manually. The build uses the manual copy at `/usr/local/maven3.9`, so the `dnf` copy is not needed.
- **File contents:** the tomcat service file, `application.properties` and `vproapp.conf` are shown as file contents. Create them with `sudo vi <path>` or `sudo nano <path>`.
- **Passwords:** `root`, `admin123`, `test` and `admin_vp` are lab values. Use secrets management for anything real.